

Analysis and Improvements of Accurate Dynamic Current Mirrors

GEORGE WEGMANN, STUDENT MEMBER, IEEE, AND ERIC A. VITTOZ, FELLOW, IEEE

Abstract—Possibilities of realizing highly accurate dynamic current mirrors based on dynamic analog techniques and improvements thereof are considered. These current mirrors, which memorize the input current, are insensitive to transistor mismatch, whereas for conventional current mirrors, the accuracy is limited by the achievable matching. Different current mirror structures are presented, the interfering parameters shown, and their influence explained. The dominant transients occurring during switching are pointed out. Improvements, which allow for increases in the performance of dynamic current mirrors, are proposed and experimental results are shown.

I. INTRODUCTION

MOS transistors need no gate current to control the drain current. Dynamic analog techniques [1] exploit this property, which allows some analog information to be stored on the gate capacitor. These techniques are used to implement offset compensation in amplifiers and to ensure adequate biasing of CMOS inverters. The dynamic comparator [2] is another application in which the analog storage capability is exploited by sequentially using the same transistor for the two devices of a differential pair. The very notion of mismatch disappears, but charge injection from MOS transistors used as switches sets a new limit.

Another ubiquitous elementary building block whose precision is limited by device mismatch is the current mirror. This handicap affects current-mode A/D converters or continuous-time filters. Until now, an increase of the current accuracy was obtained by using large transistors and by overwhelming the threshold-voltage mismatch with a high gate voltage overhead.

Improved current mirrors based on a chopper technique in a bipolar technology have been proposed [3]. The drawback of this technique is that there is a high output ripple, which must be filtered out by external components.

Manuscript received October 24, 1989; revised January 22, 1990. This work was supported by the National Program PN13 of the Swiss Foundation for Scientific Research.

G. Wegmann is with the Electronics Laboratory (LEG), Swiss Federal Institute of Technology (EPFL), CH-1015 Lausanne, Switzerland.

E. A. Vittoz is with the Centre Suisse d'Electronique et de Microtechnique S. A. (CSEM), CH-2000 Neuchâtel 7, Switzerland.
IEEE Log Number 9034695.

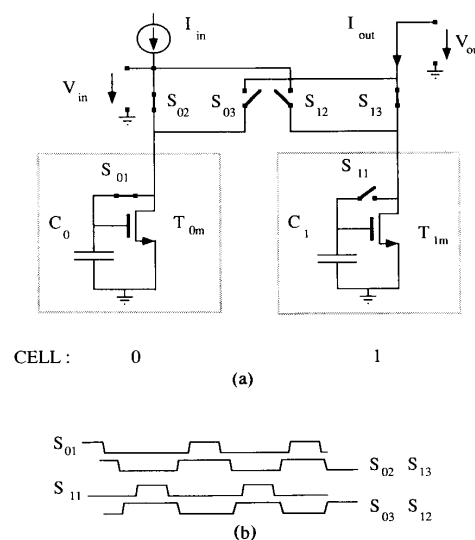


Fig. 1. (a) Principle of the dynamic current mirror built up with two identical basic cells and (b) the corresponding clock phases.

The idea of sampling and storing analog information on a capacitor can be applied to a MOS current mirror. The memorized transistor gate voltage is then an image of the imposed drain current [4]–[9]. This transistor can be sequentially connected as the input and the output device of a mirror. The current reproduced at the output depends on the accuracy of the gate voltage storage, but is independent of transistor mismatch.

II. PRINCIPLE OF A DYNAMIC CURRENT MIRROR

Fig. 1(a) shows the schematic representation of a dynamic current mirror, which is capable of memorizing and reproducing a current I_{in} imposed by an external current source [4], [5], [8], [9]. The mirror is made up of two basic cells (cells 0 and 1), which must work with complementary clocks as shown in Fig. 1(b) if a continuous output current is needed.

The mirror works as follows. While the switches S_{i1} and

S_{j2} ($j = 0, 1$) are closed, the current source feeds the input current I_{in} to the drain-gate-connected transistor T_{jm} . During this phase, switch S_{j3} is open. After opening the sampling switch S_{j1} the capacitance maintains ("memorizes") the gate voltage and thus the drain current of T_{jm} remains equal to I_{in} . When S_{j2} is open and S_{j3} is closed, the memorized drain current is available at the output. Hence T_{0m} and T_{1m} work either as independent current sources (S_{j1} open) or as drain-gate-connected transistors (S_{j1} closed).

Note that the value of the output current is independent of matching of the MOS transistors T_{jm} and of linearity or hysteresis of the capacitors C_j . Because the input voltage V_{in} for a given input current depends on the mirror transistor parameters, V_{in} varies stepwise as a function of the mismatch of the T_{jm} mirror transistors.

The performance of the dynamic current mirror is limited by the following major effects.

a) *Charge injection of the MOS transistors S_{j1} used as switches:* The mobile charge released by the inversion charge layer is shared between the drain, source, and substrate [10]. Therefore, a part of the charge released by sampling switch S_{j1} is added to the charge already stored on C_j , and the resulting error voltage ΔV alters the drain current.

b) *Transients occurring when switching S_{j2} and S_{j3} :* As long as there is a voltage difference at these switches, transients ("glitches") always exist.

c) *Junction leakage currents in the sampling switch S_{j1} :* They discharge storage capacitor C_j and hence determine the minimum switching frequency of the mirror for a given accuracy.

d) *Output conductances:* As in a classical current mirror, the reproduced output current should not depend on the output voltage. Therefore the finite output conductance of T_{jm} has to be minimized.

III. STRUCTURES AND VARIATIONS

In order to reduce the transients and the effect of the output conductance, the voltage difference applied to S_{j2} and S_{j3} should be as small as possible. One way of achieving this is to implement a cascoded structure by means of a common-gate transistor.

The simplest possible dynamic current mirror can be built by adding to the input a diode-connected transistor, which biases the output common-gate transistor, leading to a so-called "stacked mirror" [5], [8]. The drawback of this structure is its relatively high saturation voltage.

To increase the output dynamic of the current mirror, S_{j1} can be connected to the external current source I_{in} as shown in Fig. 2. But to make this mirror work correctly, T_{jm} must be prevented from coming out of saturation. The gate voltage drop on T_{0c} is included in the gate voltage of T_{jm} . Transistor T_{jm} may only remain saturated if T_{0c} is located in a separate well. If T_{0c} remains in weak inversion, this mirror works without any restrictions. But if T_{0c} leaves weak inversion, the transconductance pa-

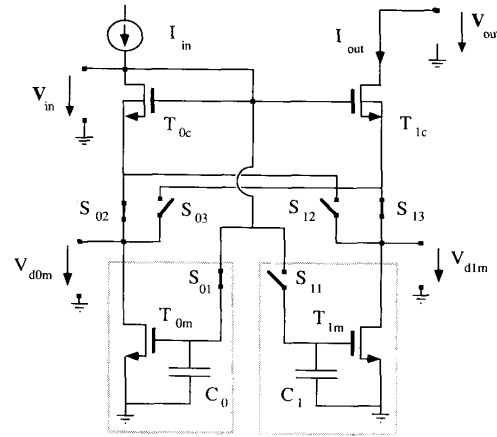


Fig. 2. Self-biased dynamic current mirror of ratio 1:1 (two basic cells) with the sampling switch S_{j1} connected directly to the input V_{in} .

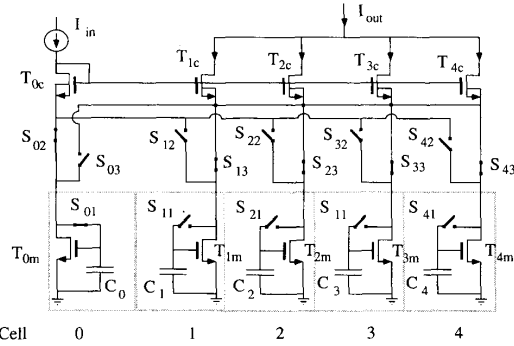


Fig. 3. Self-biased "stacked" dynamic mirror of ratio 1:4 built with five basic cells. Cell 0 memorizes the input current I_{in} .

rameters must satisfy the following condition:

$$\sqrt{\frac{\beta_m}{\beta_c}} \leq \frac{\gamma}{\gamma + 2\sqrt{2\Phi_f + V_p}} \quad (1)$$

where γ is the usual body-effect parameter, Φ_f is the Fermi potential, and V_p is the pinch-off voltage.

The elementary cell consisting of transistor T_{jm} , storage capacitor C_j , and sampling switch S_{j1} can be reproduced at will to obtain an array of current sources, each of value I_{in} [5], [8]. Some of them can be connected in parallel to get the desired nonunity integer ratios. Fig. 3 represents a self-biased "stacked dynamic current mirror" of ration 1:4 using five identical basic cells. Switches S_{j2} and S_{j3} serve to connect cell j either to the input or the output. The source nodes of the common-gate transistors are connected together to reduce the number of connections. Four cells are always connected simultaneously to the output delivering the desired current.

Ratios smaller than unity can be obtained by using a convergent algorithm [11].

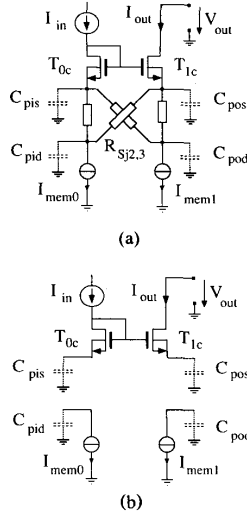


Fig. 4. Circuit configuration with: (a) overlapping clocks for S_{j2} and S_{j3} , and (b) nonoverlapping clocks for S_{j2} and S_{j3} .

For certain applications, it is possible to work at constant drain voltages, which cancels the effect of the finite output conductance. This can be achieved with the help of an amplifier [12].

A small voltage difference $|V_{d0m} - V_{d1m}|$ at switches S_{j2} and S_{j3} always exists, because of the transistor mismatch in the common-gate structure, the resistance mismatch of S_{j2} and S_{j3} , and the finite output conductances. This causes glitches as will be seen in the following section.

IV. DYNAMIC BEHAVIOR AND TRANSIENT ANALYSIS

Dynamic current mirrors are time-variant circuits. This means that a different circuit configuration is present at each switching event. The analysis of mirrors providing nonunity ratios can be reduced to one with a 1:1 ratio, because only two basic cells are interfering while switching.

For continuous-time applications the amplitude and the main time constant of the transients (glitches) are very important. Their importance is limited when the currents must only be available in a specific time window.

A. Influence of Clock Delay

It seems obvious that sampling switch S_{j1} must only be closed when S_{j2} is closed, according to the representation in Fig. 1(b), lest the gate voltage memorized on T_{jm} not be an image of the input current I_{in} .

Until now we have assumed that S_{j2} and S_{j3} can be switched simultaneously. But this is practically impossible to perform. The influence of overlapping or nonoverlapping clocks on the circuit configuration is shown in Fig. 4.

If the clocks overlap, S_{j2} and S_{j3} are all closed at the same time and the internal nodes are connected together (Fig. 4(a)). A new current mirror is formed with T_{0c} and T_{1c} . The internal parasitic capacitances C_{pxx} are charged by currents provided by this new current mirror and are discharged by the currents memorized in the basic cells [13]. Hence, the internal node voltage varies proportionally to the mismatch of T_{0c} and T_{1c} .

In the case of nonoverlapping clocks, S_{j2} and S_{j3} are all open at the same time and the different parts of the mirror are disconnected from each other (Fig. 4(b)). As a consequence, parasitic capacitances C_{pis} and C_{pos} are charged by the input current, whereas C_{pid} and C_{pod} are discharged by the currents memorized in the basic cells. Even if the time spent in such a configuration is small, large transients and nonlinear effects occur and important ac components are added; the transistors may temporarily come out of saturation, and the dc accuracy is completely spoiled [13]. Hence, this is the worst-case situation and must absolutely be avoided.

Therefore, special attention has to be paid to the design of the clock phases, as the amplitude of the transients is increased by nonsynchronous switching of S_{j2} and S_{j3} .

B. Transient Analysis

To analyze the dynamic current mirror, several assumptions have to be made, otherwise the order of the circuit is too high and no significant results can be obtained in a straightforward manner. The following assumptions are usually accepted:

- the functional capacitances C_j are much larger than the parasitic capacitances C_{pxx} ; and
- the switch resistances $R_{Sj2,3}$ are much smaller than the transconductances g_{mi} .

Hence

$$C_j \gg C_{pxx}$$

and

$$R_{Sj2,3} \cdot g_{mi} \ll 1. \quad (2)$$

The switch resistances $R_{Sj2,3}$ of S_{j2} and S_{j3} have to be small in order to avoid an excessive voltage drop on the switches, to minimize the effect of switch-resistance mismatch, and to reduce their noise contribution. The time constants associated with C_{pxx} and with $R_{Sj2,3}$ are small compared to the other ones in the circuit. Therefore, the ratio between the time constants is large enough so that their influence on the circuit can be treated separately. Thus the analytical equations can be solved.

Furthermore the variations ΔV_{gs} of the gate-to-source voltage are assumed to be small enough to allow a small-signal analysis. The effect of the voltage difference $(V_{d1m} - V_{d0m})$ and of charge injection due to S_{j2} and S_{j3} can be taken into account by modifying the initial conditions.

After switching, the voltages on both sides of the closed switches are balanced and the resulting node voltages depend on the parasitic capacitance ratio.

The transients occurring at the output can be approximated by

$$\Delta I_{out}(t) = g_{mc} \frac{C_{pod}}{C_{pod} + C_{pos}} (V_{d1m} - V_{d0m}) e^{(-t/\tau_{m0})} \quad (3)$$

where the main time constant is $\tau_{m0} = C_{pod} + C_{pos}/g_{mc}$, C_{pos} and C_{pod} are the parasitic capacitances of the output cell (Fig. 4), and g_{mc} stands for the transconductance of the common-gate transistor. The output time constant due to external loads has not been taken into account.

The average dc-current error due to glitches can be found by integrating (3), which leads to

$$\begin{aligned} \overline{\Delta I_{out}} &= \frac{1}{T_{switch}} \int_0^{T_{switch}} I_{out}(t) dt \\ &= \frac{C_{pod}(V_{d1m} - V_{d0m})}{T_{switch}} \\ &\approx f_{switch} C_{pod} \Delta V_T \end{aligned} \quad (4)$$

where f_{switch} is the switching frequency and ΔV_T is the threshold-voltage mismatch of T_{jc} and T_{0c} . The average output current error $\overline{\Delta I_{out}}$ depends linearly on the switching frequency f_{switch} and is independent of the input current, because the current-dependent term g_{mc} of (3) is cancelled out by $1/\tau_{m0}$ after integration.

The influence of the different critical switching situations and of the main time constant τ_{m0} of the output cell have been verified by simulation [13] with the circuit simulator ESACAP [14]. It has to be pointed out that $(V_{d1m} - V_{d0m})$ may remain of the same sign during a whole switching cycle. The transients then have the same polarity and do not cancel each other out as shown in (4).

If a self-biased structure is used, the variations of the input voltage V_{in} due to the mismatch of T_{jm} are transmitted to the output through the common-gate transistor T_{jc} . The amplitude of the glitches is increased, but they do not affect the dc accuracy, as their sum equals zero during a cycle.

C. Speed–Accuracy Trade-Off

A relation between the relative current error due to charge injection and the time available for the current memorization, the main time constant of the input cell τ_{mi} , can be found. The sampling switch conductance g_{onS1} can be expressed as

$$\begin{aligned} g_{onS1} &= \frac{\mu C_{ox} W_{eff}}{L_{eff}} (V_{gs} - V_T) \\ &= \frac{\mu Q_{S1}}{L_{eff}^2} \end{aligned} \quad (5)$$

where L_{eff} and W_{eff} are the effective transistor channel

length and width, respectively, μ is the mobility, C_{ox} is the capacitance per unit area, V_{gs} is the applied gate-to-source voltage, V_T is the threshold voltage, and Q_{S1} is the total channel charge of the sampling switch S_{j1} .

The residual charge injection after compensation is $\Delta Q_{inj} = \alpha Q_{S1}$ and induces an error voltage ΔV on the storage capacitor C_j , hence an output current error ΔI_{error} , which can be written as

$$\begin{aligned} \frac{\Delta I_{error}}{I_{in}} &= \frac{g_{mm} \Delta V}{I_{in}} \\ &= \frac{g_{mm} \alpha L_{eff}^2 g_{onS1}}{I_{in} \mu C_j}. \end{aligned} \quad (6)$$

The current error can be expressed as a function of the memorization time constant τ_{mi} , the principal time constant of the input cell. If g_{onS1} is high, the main time constant is given by $\tau_{mi} = C_j/g_{mm}$. If the switch conductance is small, an overshoot of the gate voltage appears and the main time constant, corresponding to the envelop, can be defined as $\tau_{mi} = 2C_{pi}/g_{onS1}$, where $C_{pi} = C_{pid} + C_{pis}$ stands for the sum of the parasitic capacitances of the internal nodes of the input cell.

Therefore, the current error can be expressed as a function of τ_{mi} for the two cases mentioned above, and the following relations are found:

$$\begin{aligned} \Delta I_{error} &= \frac{\alpha L_{eff}^2 g_{onS1}}{\mu \tau_{mi}}, \\ \text{if } g_{onS1} \gg g_{mm} \quad \tau_{mi} &= \frac{C_j}{g_{mm}} \end{aligned} \quad (7a)$$

$$\begin{aligned} \Delta I_{error} &= \frac{2g_{mm} \alpha L_{eff}^2 C_{pi}}{\mu C_j \tau_{mi}}, \\ \text{if } g_{mm} \gg g_{onS1} \quad \tau_{mi} &= \frac{2C_{pi}}{g_{onS1}}. \end{aligned} \quad (7b)$$

These relations show the trade-off between the achievable accuracy and the memorization time constant τ_{mi} .

V. INTERFERING PARAMETERS AND IMPROVEMENTS

The use of a cascoded mirror decreases the output conductance of the mirror by the cascode gain. The drawback of using multiple cascodes is a reduced output dynamic and increased leakage currents due to the increase of the number of p-n junctions in the current path. This is a limitation mainly for low currents.

Equation (6) shows the different parameters influencing the relative current error $\Delta I_{error}/I_{in}$ due to charge injection. A large gate voltage overhead $(V_{gs} - V_T)$ reduces the term g_{mm}/I_{in} , but increases the saturation voltage of the current mirror. The transconductance g_{mm} can be reduced by using a mirror structure with a modi-

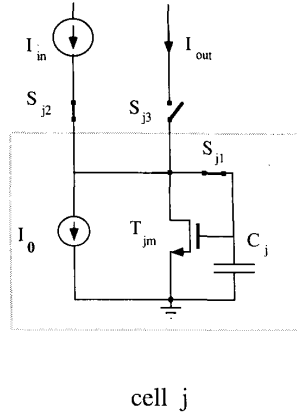


Fig. 5. Improved basic cell of a dynamic mirror, which memorizes the current difference ($I_{in} - I_0$).

fied basic cell, which is shown in Fig. 5 [7]. This improved dynamic current mirror, which memorizes only a current difference ($I_{in} - I_0$), is less sensitive to an error voltage ΔV on C_j , because for the same gate-to-source voltage V_{gs} , the transconductance $g_{mm\Delta I}$ of such an improved basic cell is reduced to

$$g_{mm\Delta I} = \frac{\Delta(I_{in} - I_0)}{I_{in}} g_{mm} \quad (8)$$

But to make this cell work correctly, the current difference $\Delta(I_{in} - I_0)$ must be larger than zero. An additional current source I_0 must be available and the memorized gate voltage of T_{jm} must be higher than the saturation voltage of this current source I_0 . Variations of input voltage V_{in} of an improved basic cell are increased in comparison to the variations of a normal one.

Different strategies can be used to reduce the effect of charge injection by sampling switch S_{j1} [10], the parameter α of (6) and (7). It is possible to simply increase the storage capacitor C_j . Another possible strategy is to achieve equipartition of the channel charge. With the help of "dummy switches," the charge released by S_{j1} can then be compensated, hence the term α is reduced. For minimal-length MOS switches the overlap capacitances are of the same importance as the gate capacitance. "Symmetrical dummy switches" as represented in Fig. 6 lower the effect of asymmetrical overlap capacitances. The charge of the overlap capacitances designated by the same capital letter cancel each other out, because they are driven by complementary clocks. Compensation accuracy is limited to the mismatch of β and V_T between S_{j1} and its "symmetrical dummy switches."

To reduce the effect of charge injection even more, the amount of channel charge in the sampling switch can be minimized by a low gate voltage overhead (Fig. 7) [9]. The idea is to reproduce the source and drain voltage of S_{j1} , which is equal to the gate voltage of T_{jm} , with the help of an image current I'_{in} and a drain-gate-connected transis-

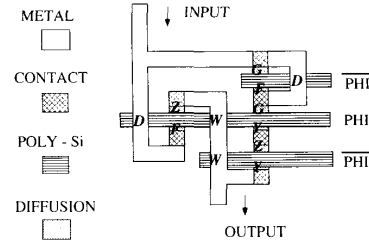


Fig. 6. Layout of the switch S_{j1} with its "symmetrical dummy switches" reducing the effect of asymmetrical overlap capacitances. The overlap capacitances named with the same capital letter have the same value, and are driven by complementary clocks.

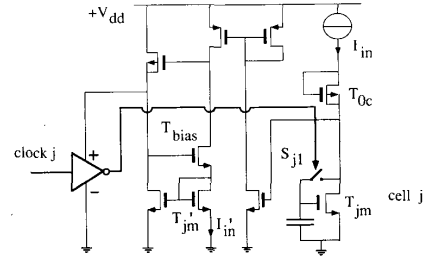


Fig. 7. Circuit minimizing the sampling switch charge and adapting the gate on-voltage of S_{j1} to the stored gate voltage of T_{jm} .

tor T'_{jm} , which is similar to the one of the basic cell. The source node of transistor T_{bias} is connected to T'_{jm} , and the gate-to-source voltage of T_{bias} is equal to that of S_{j1} . The conductance g_{onS1} varies with respect to the image current I'_{in} and the device parameters of transistor T_{bias} . Hence the charge released by S_{j1} can be controlled and minimized. Furthermore, a reduction of the gate-on voltage favors equipartition of mobile carriers forming the channel [10].

Another strategy is to suppress charge injection by negative feedback [15], with the drawback that a supplementary clock cycle is needed and that the input voltage variations are increased.

White noise contributed by S_{j2} and S_{j3} is negligible and the noise of the common-source transistors T_{jc} is divided by the cascode gain. The only white-noise contribution comes from the transistor T_{jm} connected to the output. If the sampling frequency f_{switch} is smaller than half the equivalent bandwidth $B_{eq} = 1/4\tau_{mi}$ of the sample-and-hold circuit, white noise is undersampled and the power spectral density of the white-noise component in the baseband is increased because of additional foldover terms [16]. After integrating this power spectral density, a random error component ΔV_{error} of variance kT/C_j is found. This error can only be reduced by increasing storage capacitor C_j .

$1/f$ noise is cancelled at low frequencies due to the inherent autozero technique, which introduces a double zero at the origin. But foldover terms of $1/f$ noise contribute to the increase of the white-noise component in the baseband [16].

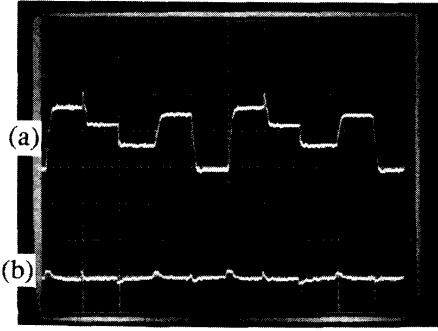


Fig. 8. Photograph of waveforms of (a) V_{in} and (b) I_{out} for an n-type current mirror of ratio 1:4 and $I_{in} = 1 \mu A$, while all phases work correctly. Time axis: $\Delta t = 200 \mu s/div$. Y axis: (a) $V_{in} = 5 mV/div$, (b) $I_{out} = 5 nA/div$.

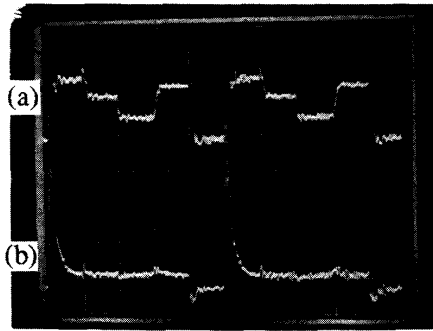


Fig. 9. Photograph of (a) V_{in} and (b) I_{out} for an n-type current mirror of ratio 1:4 and $I_{in} = 1 \mu A$, while one phase is delayed. Time axis: $\Delta t = 200 \mu s/div$. Y axis: (a) $V_{in} = 5 mV/div$, (b) $I_{out} = 5 nA/div$.

VI. EXPERIMENTAL RESULTS

Dynamic n-type and p-type current mirrors of different structures and current ratios were integrated in a $3\text{-}\mu m$ p-well CMOS technology with self-aligned contacts (SACMOS) [17]. The circuits operate either with one external phase per basic cell or even with only one external phase per mirror.

Figs. 8 and 9 show the measured variations of the input voltage V_{in} and of the output current I_{out} of the circuit shown in Fig. 3, a self-biased n-type current mirror of ratio 1:4. In Fig. 8, all clock phases work correctly, whereas in Fig. 9 the clock of one basic cell is artificially delayed to show the influence of different switching configurations (Section IV-A). Due to the mismatch of the five mirror transistors T_{jm} , the input voltage V_{in} (Figs. 8(a) and 9(a)) varies stepwise. In Fig. 8(b) the variations of the output current I_{out} are smaller than 1250 ppm, corresponding to one division. In Fig. 9(b) a large glitch can be observed, which is due to the all-switches-open configuration.

The photograph of Fig. 10 represents the main time constant of the output cell $\tau_{m0} = C_{pod} + C_{pos}/g_{mc}$, which is explicated by representing the waveform of the internal output node V_{d1m} of a current mirror of ratio 1:1 for two

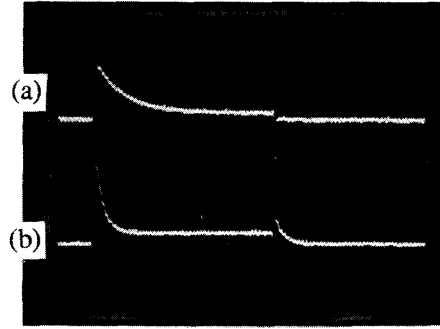


Fig. 10. Photograph clearly showing the main time constant $\tau_{m0} = C_{pod}/g_{mc}$ of the output cell for two different input currents. X axis: time = $100 \mu s/div$, (a) $I_{in} = 0.2 \mu A$. Y axis: (a), (b) $V_{d1m}(t) = 10 mV/div$, (b) $I_{in} = 0.8 \mu A$.

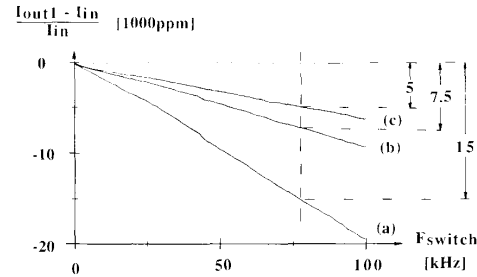


Fig. 11. Measured error $\epsilon = I_{out} - I_{in} / I_{in}$ (1000 ppm) for a self-biased n-type mirror of ratio 1:1 as a function of the switching frequency f_{switch} for three different input currents I_{in} according to (4): (a) $I_{in} = 0.25 \mu A$, (b) $I_{in} = 0.5 \mu A$, and (c) $I_{in} = 0.75 \mu A$.

different current ratios. The internal node capacitance C_{pos} is artificially increased by that of the probe and the clocks are delayed to provide a better representation. The currents are $I_{in} = 0.2 \mu A$ (Fig. 10(a)) and $I_{in} = 0.8 \mu A$ (Fig. 10(b)). As the cascode transistor works in weak inversion, the g_{mc} is proportional to I_{in} , and thus to $1/\tau_{m0}$. It has to be pointed out that the glitches in Fig. 10(b) are of the same polarity; hence they do not cancel each other out and the dc-current error depends on the switching frequency.

In Fig. 11 this linear dependence is shown. It is due to the linear increase of the number of glitches per time period. According to (4) the error due to the glitches is constant for the three different current values, because the common-gate transistor remains in weak inversion and the clock configuration remains constant during the measurement. The slope of these graphs or even its sign depends on the clock delays and on the mismatch between the diode-connected transistor and the common-gate transistor.

For very low switching frequencies, the current error is increased (not visible on this figure), because the storage capacitor is discharged by the leakage currents of the sampling switch junctions.

Fig. 12 represents the measured error of a self-biased n-type current mirror with two outputs of ratios 1:1 and

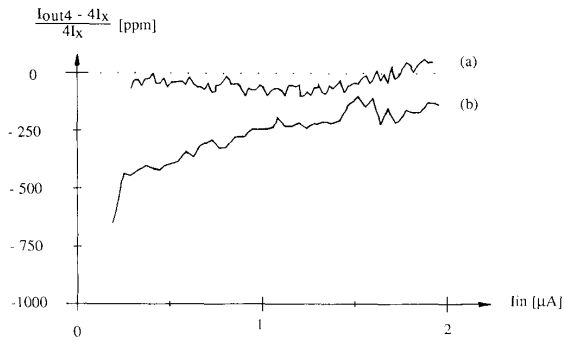


Fig. 12. Measured error $\epsilon = (I_{out4} - 4I_x) / 4I_x$ (ppm) for a self-biased n-type mirror with $I_{in} : I_{out1} : I_{out4} = 1 : 1 : 4$ as a function of the input current I_{in} with $V_{out} = 3$ V and $f_{switch} = 1$ kHz. (a) $I_x = I_{out1}$. (b) $I_x = I_{in}$.

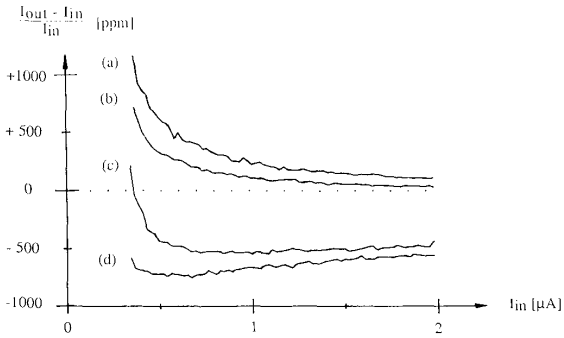


Fig. 13. Measured error $\epsilon = (I_{out} - I_{in}) / I_{in}$ (ppm) for n-type mirror of ratio 1:1:1 as a function of the input current I_{in} with (a), (c) $V_{out} = 3$ V and (b), (d) $V_{out} = 2$ V. $f_{switch} = 1$ kHz. (a), (b) Improved basic cell memorizing a current difference (ratio $I_0 / I_{in} = 0.85$). (c), (d) Normal basic cell.

1:4, which was obtained by repeating six elementary cells [5], [8]. During the measurement, the output voltage V_{out} was equal to 3 V and the switching frequency f_{switch} was 1 kHz. Curve (a) shows the error of input current I_{in} compared to a four times larger output current I_{out} , whereas curve (b) compares two output currents in a 1:4 ratio. The accuracy is better when two output currents are compared to each other than if an output current is compared to an input current, because first-order errors cancel each other out.

Fig. 13 compares the error of a dynamic n-type mirror built with a normal basic cell (curves (c) and (d)) to that of a mirror designed with an improved basic cell (curves (a) and (b)), as shown in Fig. 5. The different kinds of current mirrors are built with the same mirror transistors, and the mirror configurations are obtained with the help of auxiliary switches. So the comparison of the different kinds of mirrors is valid. The current ratio I_{in} and I_0 is $I_0 / I_{in} = 0.85$ (Fig. 5) and the output voltage is constant during the measurement ($V_{out} = 3$ V for curves (a) and (c) in Fig. 10, and $V_{out} = 2$ V for curves (b) and (d)). The output conductance is not affected by using either the normal basic cell or the improved one.

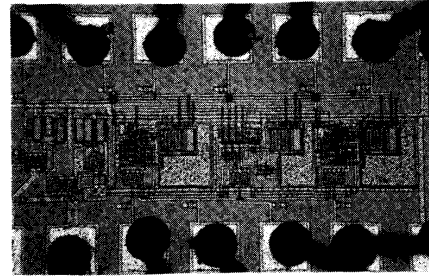


Fig. 14. Photograph of dynamic current mirror of ratio 1:1:1.

Fig. 14 shows a photograph of a dynamic current mirror with two outputs, which is built with three basic cells. This mirror can work either as a normal basic cell or with an improved one. The storage capacitors C_f are equal to 3 pF and, since the mirror dc accuracy does not depend on mismatch, they can have different shapes.

VII. CONCLUSIONS

The basic principles of different accurate dynamic current mirrors based on dynamic analog techniques have been presented and the interfering parameters, such as charge injection, clock overlapping, output conductance, noise, and leakage currents have been analyzed. The proposed dynamic current mirrors require no trimming and are insensitive to the matching or the linearity of the capacitances or transistors, and thus to process variations. The transients, which may occur during a cycle, have been analyzed and their origin shown. The trade-off between accuracy and memorization time constant has been specified. Experimental results of different kinds of dynamic current mirrors have been presented, which confirm the influence of the different parameters. The accuracy obtained makes these current mirrors useful for different applications needing accurate current mirrors, such as D/A [7] or A/D [12], [18], [19] converters, continuous-time filters [20], neural networks, or switched current filters [21].

REFERENCES

- [1] E. A. Vittoz, "Dynamic analog techniques," in *VLSI Circuits for Telecommunications*, Y. P. Tsividis and P. Antognetti, Eds. Englewood Cliffs, NJ: Prentice Hall, 1985.
- [2] Y. S. Yee *et al.*, "A 1mV CMOS comparator," *IEEE J. Solid-State Circuits*, vol. SC-13, pp. 294-297, June 1978.
- [3] R. J. Van de Plassche, "Dynamic element matching for high accuracy monolithic D/A converters," *IEEE J. Solid-State Circuits*, vol. SC-11, no. 6, pp. 795-800, Dec. 1976.
- [4] H. Oguey, private communication, 1978.
- [5] E. A. Vittoz and G. Wegmann, "High precision current mirrors," final seminar, project "CMOS functional blocks," Swiss National Research Foundation PN 13, May 1988.
- [6] S. J. Daubert, D. Vallancourt, and Y. P. Tsividis, "Current copier cell," *Electron. Lett.*, vol. 24, pp. 1560-1562, Dec. 8, 1988.

- [7] W. Groeneveld, H. Schouwenaars, and H. Termeer, "A self calibration technique for monolithic high-resolution D/A converters," in *ISSCC Dig. Tech. Papers* (New York), 1989, pp. 22-23.
- [8] G. Wegmann and E. A. Vittoz, "Very accurate dynamic current mirrors," *Electron. Lett.*, vol. 25, pp. 644-646, May 11, 1989.
- [9] G. Wegmann and E. A. Vittoz, "Analysis and improvements of highly accurate dynamic current mirrors," in *Proc. ESSCIRC* (Vienna), Sept. 1989.
- [10] G. Wegmann, E. A. Vittoz, and F. Rahali, "Charge injection in analog MOS switches," *IEEE J. Solid-State Circuits*, vol. SC-22, no. 6, pp. 1091-1097, Dec. 1987.
- [11] J. Robert, P. Deval, and G. Wegmann, "Very accurate current divider," *Electron. Lett.*, vol. 25, pp. 912-913, July 6, 1989.
- [12] D. G. Nairn and C. A. T. Salama, "Current mode analog to digital converters," in *Proc. ISCAS*, May 1989, pp. 1588-1591.
- [13] G. Wegmann and E. A. Vittoz, "Basic principles of accurate dynamic current mirrors," to be published in *Proc. IEE G, Circuits, Devices and Systems*.
- [14] P. Stangerup, "ESACAP—A PC-implemented general purpose circuit simulator," *IEEE Circuits and Devices Magazine*, July 1988.
- [15] H. P. Lic, "Switched capacitor feedback sample-and-hold circuit," U.S. Patent 4 585 956, Apr. 1986.
- [16] C. C. Enz, "High precision CMOS micropower amplifiers," Ph.D. dissertation, no. 802, Swiss Federal Inst. of Technology (EPFL), Lausanne, Switzerland, 1989.
- [17] R. E. Lüscher and J. Solo de Zaldivar, "A new approach to a high density CMOS process: SACMOS," in *ISSCC Dig. Tech. Papers* (New York), 1985, pp. 260-261.
- [18] J. Robert, P. Deval, and G. Wegmann, "Novel CMOS pipelined A/D converter architecture using current mirrors," *Electron. Lett.*, vol. 25, no. 11, pp. 691-692, May 25, 1989.
- [19] P. Deval, G. Wegmann, and J. Robert, "CMOS pipelined A/D converter using current dividers," *Electron. Lett.*, vol. 25, no. 20, Sept. 28, 1989.
- [20] F. Krummenacher, private communication.
- [21] J. B. Hughes, N. C. Bird, I. C. Macbeth, "Switched currents—A new technique for analogue sampled-data signal processing," in *Proc. ISCAS*, (Portland), 1989, pp. 1584-1587.



George Wegmann (S'82) was born in Bergamo, Italy, on May 20, 1960. He received the M.S. degree in electrical engineering from the Swiss Federal Institute of Technology (EPFL), Lausanne, Switzerland, in 1985.

Since 1985 he has been a Research Assistant at EPFL, where he is currently working towards the Ph.D. degree on the subject of micropower analog CMOS circuits.



Eric A. Vittoz (A'63-M'72-SM'87-F'89) was born in Lausanne, Switzerland, on May 9, 1938. He received the M.S. and the Ph.D. degrees in electrical engineering from the Swiss Federal Institute of Technology (EPFL), Lausanne, Switzerland, in 1961 and 1969, respectively.

After spending one year as a Research Assistant, he joined the Centre Electronique Horloger SA (CEH), Neuchâtel, Switzerland, in 1962, where he became involved in micropower integrated circuit developments for watches, while preparing a thesis in the same field. In 1971 he was appointed Vice-Director of CEH supervising advanced developments in electronic watches and other micropower systems. Since 1984 he has been a Director of the Swiss Center for Electronics and Microtechnology (CSEM) in Neuchâtel, in charge of the Circuit and System Research Division. His field of personal research is the design of low-power analog circuits in CMOS technologies, with a recent emphasis on their application to neural networks. Since 1975 he has also been lecturing and supervising student projects in analog circuit design at EPFL, where he became a Professor in 1982. He has published more than 60 technical papers and holds 23 patents.